# Customer Shopping Behavior Analysis

## 1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

## 2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping Behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

## 3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported dataset using pandas.
- **Initial Exploration:** Used df.info()to check structure and .describe()for summary statistics.

| | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discou Appli |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 3900.000000 | 3900.000000 | 3900 | 3900 | 3900 | 3900.000000 | 3900 | 3900 | 3900 | 3900 | 3863.000000 | 3900 | 3900 | 39 |
| unique | NaN | NaN | 2 | 25 | 4 | NaN | 50 | 4 | 25 | 4 | NaN | 2 | 6 | |
| top | NaN | NaN | Male | Blouse | Clothing | NaN | Montana | M | Olive | Spring | NaN | No | Free Shipping | |
| freq | NaN | NaN | 2652 | 171 | 1737 | NaN | 96 | 1755 | 177 | 999 | NaN | 2847 | 675 | 22 |
| mean | 1950.500000 | 44.068462 | NaN | NaN | NaN | 59.764359 | NaN | NaN | NaN | NaN | 3.750065 | NaN | NaN | Na |
| std | 1125.977353 | 15.207589 | NaN | NaN | NaN | 23.685392 | NaN | NaN | NaN | NaN | 0.716983 | NaN | NaN | Na |
| min | 1.000000 | 18.000000 | NaN | NaN | NaN | 20.000000 | NaN | NaN | NaN | NaN | 2.500000 | NaN | NaN | Na |
| 25% | 975.750000 | 31.000000 | NaN | NaN | NaN | 39.000000 | NaN | NaN | NaN | NaN | 3.100000 | NaN | NaN | Na |
| 50% | 1950.500000 | 44.000000 | NaN | NaN | NaN | 60.000000 | NaN | NaN | NaN | NaN | 3.800000 | NaN | NaN | Na |
| 75% | 2925.250000 | 57.000000 | NaN | NaN | NaN | 81.000000 | NaN | NaN | NaN | NaN | 4.400000 | NaN | NaN | Na |
| max | 3900.000000 | 70.000000 | NaN | NaN | NaN | 100.000000 | NaN | NaN | NaN | NaN | 5.000000 | NaN | NaN | Na |

| Discount Applied | Promo Code Used | Previous Purchases | Payment Method | Frequency of Purchases |
|---|---|---|---|---|
| 3900 | 3900 | 3900.000000 | 3900 | 3900 |
| 2 | 2 | NaN | 6 | 7 |
| No | No | NaN | PayPal | Every 3 Months |
| 2223 | 2223 | NaN | 677 | 584 |
| NaN | NaN | 25.351538 | NaN | NaN |
| NaN | NaN | 14.447125 | NaN | NaN |
| NaN | NaN | 1.000000 | NaN | NaN |
| NaN | NaN | 13.000000 | NaN | NaN |
| NaN | NaN | 25.000000 | NaN | NaN |
| NaN | NaN | 38.000000 | NaN | NaN |
| NaN | NaN | 50.000000 | NaN | NaN |

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Ratingcolumn using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
    - Created **age_group** column by binning customer ages.
    - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if discount_appliedand promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

## 4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

| | gender text | revenue numeric |
|---|---|---|
| 1 | Female | 75191 |
| 2 | Male | 157890 |

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

| | customer_id bigint | purchase_amount bigint |
|---|---|---|
| 1 | 2 | 64 |
| 2 | 3 | 73 |
| 3 | 4 | 90 |
| 4 | 7 | 85 |
| 5 | 9 | 97 |
| 6 | 12 | 68 |
| 7 | 13 | 72 |
| 8 | 16 | 81 |
| 9 | 20 | 90 |
| 10 | 22 | 62 |
| 11 | 24 | 88 |



Total rows: 839 Query complete 00:00:0

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

| | item_purchased text | Average Product Rating numeric |
|---|---|---|
| 1 | Gloves | 3.86 |
| 2 | Sandals | 3.84 |
| 3 | Boots | 3.82 |
| 4 | Hat | 3.80 |
| 5 | Skirt | 3.78 |

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

| | shipping_type text | round numeric |
|---|---|---|
| 1 | Standard | 58.46 |
| 2 | Express | 60.48 |

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

| | subscription_status text | total_customers bigint | avg_spend numeric | total_revenue numeric |
|---|---|---|---|---|
| 1 | Yes | 1053 | 59.49 | 62645.00 |
| 2 | No | 2847 | 59.87 | 170436.00 |

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

| | item_purchased text | discount_rate numeric |
|---|---|---|
| 1 | Hat | 50.00 |
| 2 | Sneakers | 49.66 |
| 3 | Coat | 49.07 |
| 4 | Sweater | 48.17 |
| 5 | Pants | 47.37 |

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

| | customer_segment text | Number of Customers bigint |
|---|---|---|
| 1 | Loyal | 3116 |
| 2 | New | 83 |
| 3 | Returning | 701 |

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

| | item_rank bigint | category text | item_purchased text | total_orders bigint |
|---|---|---|---|---|
| 1 | 1 | Accessories | Jewelry | 171 |
| 2 | 2 | Accessories | Sunglasses | 161 |
| 3 | 3 | Accessories | Belt | 161 |
| 4 | 1 | Clothing | Blouse | 171 |
| 5 | 2 | Clothing | Pants | 171 |
| 6 | 3 | Clothing | Shirt | 169 |
| 7 | 1 | Footwear | Sandals | 160 |
| 8 | 2 | Footwear | Shoes | 150 |
| 9 | 3 | Footwear | Sneakers | 145 |
| 10 | 1 | Outerwear | Jacket | 163 |
| 11 | 2 | Outerwear | Coat | 161 |

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

| | subscription_status text | repeat_buyers bigint |
|---|---|---|
| 1 | No | 2518 |
| 2 | Yes | 958 |

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

| | age_group text | total_revenue numeric |
|---|---|---|
| 1 | Young Adult | 62143 |
| 2 | Middle-aged | 59197 |
| 3 | Adult | 55978 |
| 4 | Senior | 55763 |

## 5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



## 6. Business Recommendations

- **Grow Subscriptions –** Test clear member benefits, such as exclusive offers or early access, and track subscription rate, repeat purchases, and revenue per customer.
- **Strengthen Customer Loyalty –** Use the New, Returning, and Loyal segments to tailor follow-ups and rewards; monitor repeat-purchase rate and time between purchases.
- **Optimize Discount Strategy –** Identify products frequently bought with discounts, then compare sales and profit to avoid unnecessary markdowns.
- **Improve Product Positioning –** Promote products that combine strong sales with high ratings and investigate lower-rated products before increasing promotion.
- **Refine Targeted Marketing –** Use age-group, category, and gender views to design campaign tests; compare conversion, average order value, and profit before shifting budget.



